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Disorders of Potassium Metabolism

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INTRODUCTION

Potassium disorders are some of the most frequently encountered fluid and electrolyte abnormalities in clinical medicine. Patients with disorders of potassium metabolism may be asymptomatic, or they may have symptoms ranging from mild weakness to sudden death. An abnormal serum potassium level, once verified, should be promptly addressed, but inappropriate treatment can worsen symptoms and even lead to death.

NORMAL PHYSIOLOGY OF POTASSIUM METABOLISM

Potassium Intake

Potassium is necessary for essentially all cellular functions, is present in most foods, and is excreted primarily by the kidney. The typical Western diet provides about 70 mmol of potassium daily, even though the recommended intake for people with normal renal function is closer to 120 mmol per day.[1] The gastrointestinal (GI) tract efficiently absorbs potassium, and total dietary potassium intake depends on the composition of the diet. Table 10.1 shows the potassium content of several foods high in potassium.

Potassium Distribution

After absorption from the GI tract, potassium distributes rapidly into the extracellular fluid (ECF) and intracellular fluid (ICF) compartments. Cellular potassium uptake is rapid and limits the magnitude of changes in serum potassium concentration. During potassium deficiency, shift of potassium from intracellular to extracellular compartments limits the change in extracellular potassium concentration.

Most body potassium is intracellular, with only 1% to 2% in the ECF. Potassium is the major intracellular cation, with cytosolic K^+ concentrations about 100 to 120 mmol/L. Total intracellular K^+ content is 3000 to 3500 mmol in healthy adults and is found primarily in muscle (70%), with a lesser amount in bone, red blood cells, liver, and skin (Table 10.2). The electrogenic sodium pump, Na^+,K^+-ATPase, is the primary effector of this asymmetric potassium distribution: It transports two potassium ions into cells and extrudes three sodium ions, which results in high intracellular potassium concentration (high $[K^+_i]$) and low intracellular sodium concentration (low $[Na^+_i]$). Potassium-selective ion channels are the predominant determinant of the resting membrane potential. Therefore, the intracellular/extracellular $[K^+]$ ratio largely determines the resting cell membrane potential and the intracellular electronegativity. Maintenance of this ratio and membrane potential is critical for normal nerve conduction and muscular contraction.

Under some conditions, the normal distribution of potassium between the extracellular and intracellular pools is altered (Fig. 10.1). Plasma osmolality, several hormones, and exercise are frequent causes of potassium shifts. Because the amount of extracellular K^+ relative to intracellular K^+ is low, modest net movement of K^+ into or out of the extracellular pool can result in substantial changes in extracellular K^+ concentration.

Several hormones, most prominently catecholamines, insulin, and aldosterone, have important roles in regulating serum K^+. The effect of catecholamines differs depending on which adrenergic receptor subtype they activate. Activation of β_2-adrenergic receptors stimulates Na^+,K^+-ATPase, inducing cellular potassium uptake and decreasing serum K^+, whereas α_1-adrenergic receptor activation has the opposite effect. Thus, drugs that block the β_2-adrenoreceptor tend to increase serum K^+, and those that block the α_1-adrenoreceptor tend to lower serum K^+.

Insulin has important effects on serum K^+. It activates Na^+,K^+-ATPase, directly increasing cellular K^+ uptake and decreasing serum K^+. This effect is rapid and enables insulin administration to be a component of the acute therapy of hyperkalemia. Importantly, insulin stimulates Na^+,K^+-ATPase through a mechanism that is distinct from its stimulation of glucose entry and does not involve effects on either α- or β-adrenoreceptors. Thus, the effects of insulin and β_2-adrenoceptor activation are additive. In people with diabetes, hyperglycemia resulting from either the lack of adequate insulin release or responsiveness leads to hyperosmolality, which can also induce cellular potassium shifts (see later).

Aldosterone regulates serum potassium in part by altering the distribution of potassium between ECF and ICF by enhancing cellular potassium uptake through stimulation of Na^+,K^+-ATPase.[2] Indeed, aldosterone administration can cause hypokalemia in the absence of altered potassium balance through mechanisms involving altered cellular K^+ distribution.[2,3] Pharmacologic inhibition of aldosterone action with mineralocorticoid receptor blockers (MRBs) can cause hyperkalemia.

Serum osmolality is another important factor that alters cellular potassium distribution. Hyperosmolality can cause hyperkalemia when it is the result of "effective osmoles," such as mannitol or persistent hyperglycemia in the absence of adequate insulin and/or insulin responsiveness. The likely mechanism is that the increased serum osmolality induces water movement out of the cells, decreasing cell volume and increasing intracellular $[K^+]$, which stimulates K^+-permeable channels and increases K^+ movement out of cells. Urea in patients with high blood urea nitrogen (BUN) levels is an "ineffective osmole" because it rapidly crosses plasma membranes and thus does not alter cell volume. Administering glucose to a nondiabetic patient, because it stimulates insulin secretion, can actually stimulate insulin-induced cellular potassium uptake and decrease serum K^+.

Exercise has multiple effects on K^+. Cellular K^+ release is required for repolarization of contracting skeletal muscle cells and can result in mild

TABLE 10.1 Select Foods With High Potassium Content

Food	Portion Size	K^+ (mmol)
Artichoke, boiled	1, medium	27
Avocado	1, medium	38
Banana	Medium	12
Cantaloupe, cut up	1 cup	13
Grapefruit juice	8 oz	10
Hamburger, lean	8 oz	18
Milk	8 oz	10
Orange juice	8 oz	12
Potato, baked	7 oz	22
Prunes	10	16
Raisins	⅔ cup	19
Sirloin steak	8 oz	23
Squash	1 cup	15–20
Tomato juice	6 oz	10
Tomato paste	½ cup	31

TABLE 10.2 Distribution of Total Body Potassium in Organs and Body Compartments

Organ/Fluid	Total K^+ Amount	Body Compartment	K^+ Concentration
Muscle	2650 mmol	Intracellular fluid	100–120 mmol/L
Liver	250 mmol	Extracellular fluid	~4 mmol/L
Interstitial fluid	35 mmol		
Red blood cells	350 mmol		
Plasma	15 mmol		

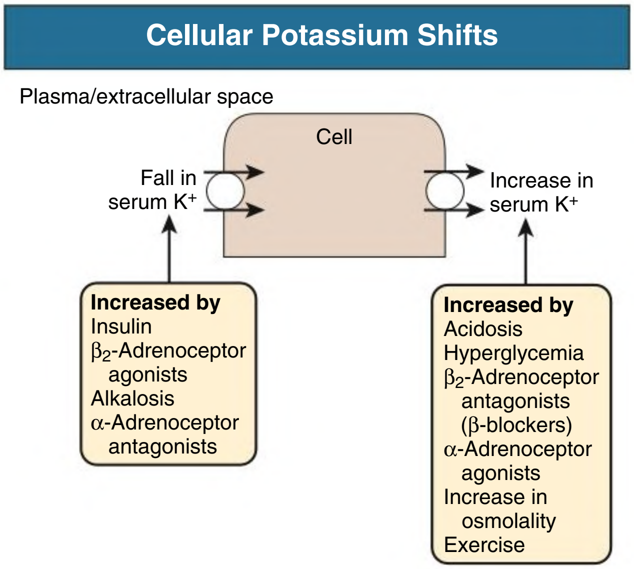

Fig. 10.1 Regulation of extracellular/intracellular potassium shifts.

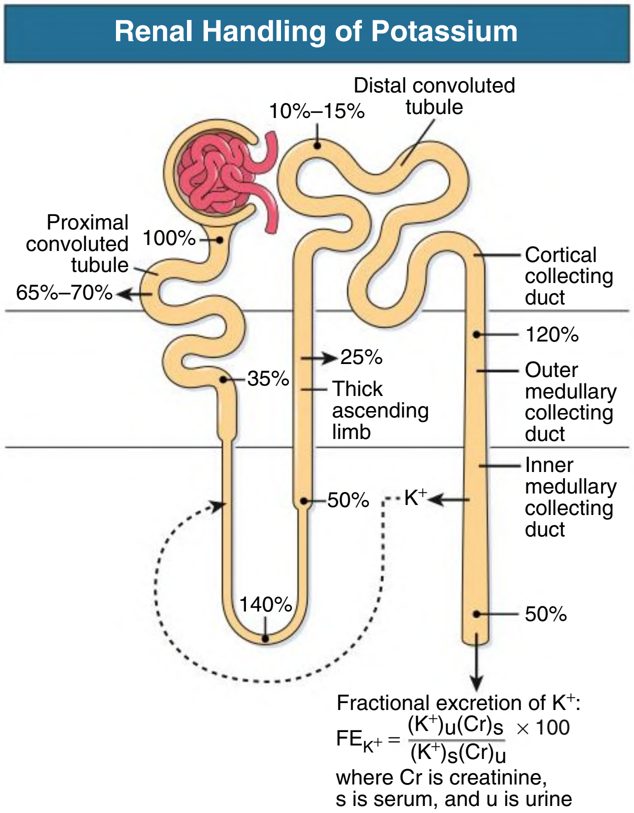

Fig. 10.2 Renal handling of potassium.

hyperkalemia during strenuous physical exercise. Catecholamines released during exercise can contribute to this hyperkalemia because α_1-adrenergic receptor activation shifts potassium out of cells. The local increase in extracellular potassium induces arterial dilation in normal blood vessels, which increases skeletal muscle blood flow and acts as an adaptive mechanism during exercise. The skeletal muscle interstitial acidosis associated with intense exercise may also contribute to K^+ release.[4] Catecholamines released during exercise, however, also activate β_2-adrenoceptors, which stimulates skeletal muscle cellular potassium uptake and minimizes the severity of exercise-induced hyperkalemia and can also lead to hypokalemia after cessation of exercise. With preexisting potassium depletion, postexercise hypokalemia may be severe and can cause rhabdomyolysis.[5]

Metabolic acidosis is often associated with abnormal serum potassium. Although hyperkalemia may be seen with lactic acidosis, this may be because of tissue ischemia leading to cellular death and cytoplasmic K^+ release. In diabetic ketoacidosis, the hyperglycemia and insulin deficiency are likely the primary mechanisms of the associated hyperkalemia. In type 4 renal tubular acidosis (RTA), hyperkalemia contributes to the metabolic acidosis by inhibiting net acid excretion in the form of ammonium.[6]

Renal Potassium Handling With Normal Kidney Function

Long-term potassium homeostasis occurs primarily through changes in renal potassium excretion. Serum potassium is almost completely ionized, is not bound to plasma proteins, and is filtered efficiently by the glomerulus (Fig. 10.2). The proximal tubule reabsorbs the majority (~65%–70%) of filtered potassium, but there is relatively little variation in proximal tubule potassium reabsorption in response to hypokalemia or hyperkalemia. In the loop of Henle, potassium is secreted in the descending loop, at least in deep nephrons, particularly with adaptation to a large K^+ intake, and is reabsorbed in the ascending loop through the action of the Na^+-K^+-$2Cl^-$ cotransporter (Fig. 10.3A). However, most K^+ transported by this protein is recycled back into the tubular lumen through an apical K^+ channel, and thus there is little net

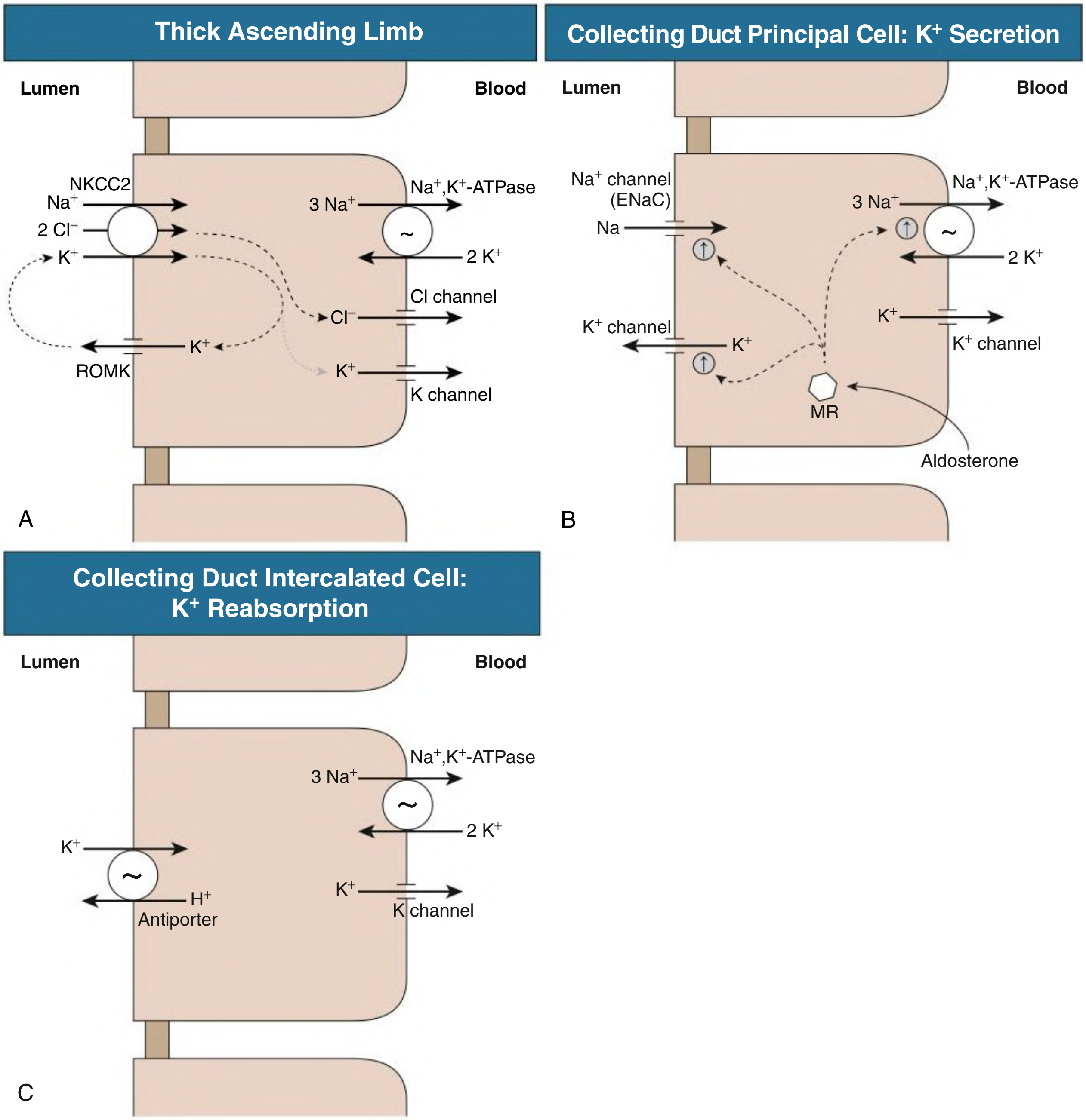

Fig. 10.3 Mechanisms of Potassium Reabsorption and Secretion. Potassium transport in the thick ascending limb of Henle loop (TAL) and in the collecting tubule principal and intercalated cells. (A) In the TAL, the majority of K^+ reabsorbed by the apical Na+,K^+-2Cl^- cotransporter (NKCC2) recycles across the apical membrane through apical renal outer medulla K^+ (ROMK) channel. (B) In the cortical collecting duct principal cell, K^+ secretion involves integrated functions of the basolateral Na^+, K^+-ATPase, apical Na^+ channel (ENaC), and apical K^+ channel. Aldosterone stimulates this process through interaction with the mineralocorticoid receptor (MR), resulting in increased expression and activity of each of these processes. Although cortisol can also activate MR, the enzyme 11β-hydroxysteroid dehydrogenase type 2, which is present in the principal cell, converts cortisol to cortisone, a steroid hormone that does not activate MR. (C) Collecting duct intercalated cells can reabsorb K^+ through the actions of an apical H^+-K^+-ATPase.

potassium reabsorption. This absorption can be reversed to secretion by administration of a loop diuretic or by substantial potassium loading. Nonetheless, the magnitude of the change in loop of Henle potassium transport in various physiologic conditions is relatively small.

The collecting duct and its proximal extension, the initial collecting tubule, are the primary sites where the kidney regulates K^+ excretion. These segments have the ability both to secrete K^+, enabling adaptation to K^+ excess states, and to reabsorb K^+, enabling adaptation to K^+ depletion states. The principal cell of the cortical collecting duct secretes potassium, whereas intercalated cells throughout the entire collecting duct reabsorb potassium. In the principal cell, sodium is reabsorbed through the apical epithelial sodium channel (ENaC), which stimulates basolateral Na^+,K^+-ATPase (see Fig. 10.3B); active potassium uptake by this protein maintains a high intracellular K^+.

After basolateral potassium uptake, K^+ is secreted into the luminal fluid by two types of potassium channels and by KCl cotransporters. Intercalated cells actively reabsorb potassium through an apical H^+-K^+-ATPase[7] (Fig. 10.3C); this protein actively secretes H^+ into the luminal fluid in exchange for reabsorption of luminal potassium. The presence of two separate potassium transport processes, secretion by principal cells and reabsorption by intercalated cells, contributes to rapid and effective regulation of renal potassium excretion.

Several factors influence principal cell potassium secretion. These include luminal flow rate, distal sodium delivery, aldosterone, extracellular potassium intake, and extracellular pH. Luminal flow rate and distal sodium delivery frequently vary together, but changes in luminal flow rate appear more important by activating flow-induced K^+ secretion through large conductance (BK) potassium channels.[8,9] Reduced luminal flow, such as occurs in prerenal azotemia and in obstructive uropathy, may contribute to the hyperkalemia that is often seen in these conditions. Decreased sodium reabsorption, whether from reduced luminal sodium delivery or treatment with ENaC inhibitors (i.e., potassium-sparing diuretics), decreases K^+ secretion by altering electrochemical forces for K^+ secretion. Conversely, increased sodium delivery to the collecting duct, as may occur with a high salt diet or with administration of either loop or thiazide diuretics, increases principal cell sodium reabsorption and causes a secondary increase in potassium secretion. Aldosterone has many effects that increase principal cell potassium secretion, including increased Na^+,K^+-ATPase, increased apical expression of ENaC, and increased apical K^+ channels. The net effect is increased principal cell-mediated K^+ secretion. Changes in extracellular potassium directly alter Na^+,K^+-ATPase activity, thereby altering K^+ secretion. Metabolic acidosis decreases K^+ secretion, both through direct effects on potassium channels and through changes in interstitial ammonia concentration, which decreases K^+ secretion.[10]

Intercalated cell-mediated potassium reabsorption occurs in parallel with principal cell-mediated potassium secretion. Active potassium reabsorption occurs through the action of the potassium-reabsorbing protein, H^+-K^+-ATPase. The major factors regulating H^+-K^+-ATPase expression and activity include potassium balance, aldosterone, and acid-base status. Potassium depletion increases H^+-K^+-ATPase expression, resulting in increased active potassium reabsorption and decreased net potassium excretion. Aldosterone increases H^+-K^+-ATPase expression and activity and, by decreasing net potassium excretion, may minimize changes in urinary potassium excretion during aldosterone excess that otherwise would lead to more severe hypokalemia. Metabolic acidosis has both direct and indirect effects, mediated through alterations in ammonia metabolism, that increase H^+-K^+-ATPase potassium reabsorption.[6]

Regulators of renal K^+ transport in the distal nephron include the "with no lysine" (WNK) kinases.[11,12] Under basal conditions, WNK kinases activate Na^+ reabsorption in the distal convoluted tubule and inhibit the renal outer medulla potassium (ROMK) channel. This combination of effects—increased DCT Na^+ reabsorption, which decreases collecting duct Na^+ delivery, in conjunction with decreased ROMK expression—promotes decreased K^+ secretion. The specific mechanisms through which WNK regulates K^+ secretion are under active investigation, and medications targeting WNK inhibition are in development and may in the future enable entirely new treatments of hypertension and K^+ disorders.[13]

The intestinal tract also contributes to renal potassium homeostasis. Potassium sensors, present in either the intestinal tract or the portal venous system, elicit a rapid increase in renal potassium excretion through mechanisms independent of serum potassium and aldosterone.[2,14] This reflex system, which is still not understood fully, supplies a mechanism to "sense" dietary potassium intake and alter renal potassium excretion without involving aldosterone concentration and prior to changes in serum potassium. This mechanism contributes to the greater safety of oral compared with intravenous (IV) potassium administration.

The contribution of the intestinal tract to K^+ excretion in fecal contents is unclear. An early study suggested stool potassium averaged 12% of dietary intake in normal subjects and 34% in patients with severe renal insufficiency.[15] In contrast, a study examining anephric patients showed fecal K^+ excretion averaged approximately 3 mEq/day, which did not differ from that observed in control patients with intact renal function.[16] In this latter study, fecal K^+ excretion was not altered by either MRB therapy or mineralocorticoid agonists.[16]

Renal Potassium Handling in Chronic Kidney Disease

Because the kidneys are the major route for elimination of potassium, as kidney function declines, the balance between dietary potassium intake, renal potassium excretion, and baseline serum potassium changes. In general, most patients with chronic kidney disease (CKD) maintain their serum potassium in the normal range, although there is a graded increase in mean serum K^+ as the glomerular filtration rate (GFR) declines. The risk of developing hyperkalemia is increased in patients with stage 4 and 5 CKD; patients with stage 3 CKD who have diabetes mellitus, tubulointerstitial disease, or receive certain drugs also are at increased risk.

Many of the medications used in the treatment of patients with CKD alter potassium homeostasis. Common medications that predispose to hyperkalemia (Table 10.3) include agents that inhibit the epithelial Na channel (ENaC) or the renin-angiotensin-aldosterone system (RAAS), nonsteroidal antiinflammatory drugs (NSAIDs), and calcineurin inhibitors. Medications that can directly inhibit ENaC, such as amiloride, triamterene, trimethoprim, and pentamidine, acutely reduce the rate of renal K^+ excretion and can cause hyperkalemia. Drugs that inhibit the RAAS, such as mineralocorticoid receptor blockers (MRBs), angiotensin receptor blockers (ARBs), and angiotensin-converting enzyme (ACE) inhibitors, direct renin inhibitors, and heparin can inhibit aldosterone's action or generation, which reduces renal K^+ excretion. Mineralocorticoid blockers are increasingly used in patients with CHF and with refractory hypertension and may slow the progression of CKD. NSAIDs that inhibit prostaglandin synthesis and β-blockers that inhibit renin release and catecholamine action can also increase the risk of hyperkalemia. In contrast, both loop and thiazide diuretics increase renal K^+ excretion and predispose to hypokalemia.

Patients with CKD generally tolerate hyperkalemia with fewer cardiac and electrocardiographic (ECG) abnormalities than do patients with normal kidney function, although the mechanism is incompletely understood. In particular, patients with CKD appear to tolerate serum [K^+] of 5.0 to 5.5 mmol/L with no significant adverse effect, and levels of 5.5 to 6.0 mmol/L are associated with lower mortality than [K^+] of 3.5 to 3.9 mmol/L.[17] Nevertheless, severe hyperkalemia (>6.0 mmol/L or presence of ECG changes) can be lethal and should be treated aggressively.

HYPOKALEMIA

Epidemiology

The incidence of potassium disorders depends greatly on the patient population. Less than 1% of adults with normal kidney function who are not receiving medications develop frank hypokalemia or hyperkalemia. However, diets with high sodium and low potassium content may lead to potassium depletion. Thus, hypokalemia or hyperkalemia suggests that either an underlying disease is present or that the

TABLE 10.3 **Drugs Classes Associated With Hyperkalemia**

Class	Mechanism	Examples
Potassium-containing drugs	Increased potassium intake	KCl, penicillin G, Na citrate, K citrate
β-Adrenergic receptor blockers (β-blockers)	Inhibit renin release	Propranolol, metoprolol, atenolol
ACE inhibitors	Inhibit conversion of angiotensin I to angiotensin II	Captopril, lisinopril
ARBs	Inhibit activation of AT_1 receptor by angiotensin II	Losartan, valsartan, irbesartan
Direct renin inhibitors	Inhibit renin activity, leading to decreased angiotensin II production	Aliskiren
Heparin	Inhibit aldosterone synthase, rate-limiting enzyme for aldosterone synthesis	Heparin sodium
Aldosterone receptor antagonists	Block aldosterone receptor activation	Spironolactone, eplerenone, finerenone
Potassium-sparing diuretics	Block collecting duct apical ENaC Na channel, decreasing gradient for K^+ secretion	Amiloride, triamterene; certain antibiotics, specifically trimethoprim and pentamidine
NSAIDs and COX-2 inhibitors	Inhibit prostaglandin stimulation of collecting duct K^+ secretion; inhibit renin release	Ibuprofen, diclofenac
Digitalis glycosides	Inhibit Na^+, K^+-ATPase necessary for collecting duct K^+ secretion and regulation of K^+ distribution into cells	Digoxin
Calcineurin inhibitors	Inhibit Na^+, K^+-ATPase necessary for collecting duct K^+ secretion	Cyclosporine, tacrolimus

ACE, Angiotensin-converting enzyme; *ARBs,* angiotensin receptor blockers; *COX,* cyclooxygenase; *ENaC,* epithelial sodium channel; *NSAIDs,* nonsteroidal antiinflammatory drugs.

individual is taking drugs that alter potassium handling. For example, hypokalemia may be present in as many as half of patients taking diuretics[18] and is present in many patients with primary or secondary hyperaldosteronism.

Clinical Manifestations

Potassium deficiency, because it alters the ratio of extracellular to intracellular potassium, alters the resting membrane potential, which can impair normal functioning of almost every cell in the body. Several studies have shown that hypokalemia, even when mild, is associated with increased long-term mortality. Overall, children and young adults tolerate hypokalemia better than elderly persons. Prompt correction is warranted in patients with coronary heart disease or in patients receiving digitalis glycosides because hypokalemia increases the risk of lethal cardiac arrhythmias.

Mortality

Several clinical studies have shown chronic hypokalemia is associated with increased mortality.[19–23] This association persists after extensive statistical analysis of confounding and associated factors. At present, whether correction of the abnormal K^+ alters the associated mortality is unknown. However, because treatment is generally well tolerated, targeting a serum/plasma K^+ of 4.0 mEq/L or greater is a reasonable goal.

Cardiovascular

Potassium deficiency increases blood pressure, salt-sensitive changes in BP, and the cardiac sensitivity to arrhythmias. Epidemiologic studies link hypokalemia and a low-potassium diet with an increased prevalence of hypertension, and experimental studies show hypokalemia increases blood pressure by 5 to 10 mm Hg and that potassium supplementation can lower blood pressure by a similar amount.[24] Potassium deficiency also increases the effect of dietary NaCl on blood pressure. This effect involves multiple mechanisms, including stimulating sodium retention and increasing intravascular volume and sensitizing the vasculature to endogenous vasoconstrictors.[24] In part, sodium retention is related to increased NCC- and ENaC-mediated sodium reabsorption in the distal convoluted tubule and cortical collecting duct, respectively.[25]

Hypokalemia also predisposes to arrhythmia, including ventricular tachycardia and ventricular fibrillation[26] and the risk of sudden cardiac death. Diuretic-induced hypokalemia is of particular concern because sudden cardiac death may occur more frequently in those treated with thiazide diuretics.[26] Ventricular arrhythmias are also more common in patients receiving digoxin who develop hypokalemia.

Hormonal

Hypokalemia impairs insulin release and induces insulin resistance, resulting in worsened glucose control in diabetic patients.[27] The insulin resistance that usually occurs with thiazide diuretic therapy is caused by endothelial dysfunction mediated by thiazide-induced hypokalemia and hyperuricemia.[28]

Muscular

Hypokalemia can lead to skeletal muscle weakness and to an increased risk of exertion-related rhabdomyolysis. Hypokalemia hyperpolarizes skeletal muscle cells, thereby impairing the ability to generate the action potential needed for muscle contraction. Hypokalemia also reduces skeletal muscle blood flow, possibly by impairing local nitric oxide release; this effect can predispose patients to rhabdomyolysis during vigorous exercise.[29] Severe hypokalemia can also lead to respiratory muscle weakness and, if severe, respiratory failure.

Kidney Related

Hypokalemia leads to several important disturbances of kidney function. Reduced medullary blood flow and increased renal vascular resistance may predispose to hypertension, tubulointerstitial and cystic changes, alterations in acid-base balance, and impairment of urinary concentrating mechanisms.

Potassium depletion causes tubulointerstitial fibrosis that is generally greatest in the outer medulla. The degree of reversibility is related to the duration of hypokalemia, and if prolonged, hypokalemia may result in kidney failure. Experimental studies suggest an increased risk for irreversible kidney injury when hypokalemia is present during the neonatal period.[30] Longstanding potassium depletion also causes kidney hypertrophy and predisposes to kidney cyst formation, particularly when there is increased mineralocorticoid activity.

Metabolic alkalosis is a common acid-base consequence of severe potassium depletion. It results primarily from increased net acid excretion caused by increased potassium reabsorption and proton excretion in the collecting tubule as well as enhanced ammonia excretion, in combination in stimulation of proximal tubule bicarbonate reabsorption.[31]

Severe hypokalemia can cause mild polyuria, typically 2 to 3 L/day. Both increased thirst and mild nephrogenic diabetes insipidus contribute to the polyuria.[32] The nephrogenic diabetes insipidus is caused by decreased expression of several proteins, such as the water transporter aquaporin 2 (AQP2), and the urea transporters (UT)-A1, UT-A3, and UT-B, which are involved in urine concentration and water reabsorption.

Hypokalemia increases both ammonia production and expression of collecting duct proteins involved in ammonia secretion. This leads to increased ammonia excretion in the urine, increasing net acid excretion and leading to development of metabolic alkalosis. In addition, there is increased ammonia to the renal veins, from where it is added to the systemic circulation. In patients with acute or chronic liver disease, this increased ammonia delivery may exceed hepatic ammonia clearance, increase plasma ammonia levels and either precipitate or worsen hepatic encephalopathy.[6]

Etiology

Hypokalemia results typically from one of four etiologies: pseudohypokalemia, redistribution, extrarenal potassium loss, or urinary potassium loss; multiple etiologies may coexist in a specific patient.

Pseudohypokalemia

Pseudohypokalemia refers to the condition in which serum potassium decreases, artifactually, after phlebotomy. The most common cause is acute leukemia; the large numbers of abnormal leukocytes take up potassium when the blood is stored in a collection vial for prolonged periods at room temperature. Rapid separation of plasma and storage at 4°C is used to confirm this diagnosis. If confirmed, only plasma potassium measurements, and not serum levels, should be used for subsequent clinical decision making to avoid this artifact leading to inappropriate treatment.

Redistribution

Because less than 2% of total body potassium is in the ECF compartment, quantitatively small potassium shifts from the ECF to the ICF compartment can cause substantial hypokalemia. A chronic increase in aldosterone secretion increases the pump-leak kinetics and reduces plasma K^+ in the absence of perceptible increases in urinary K^+ excretion if intake is constant.

A clinically rare, but dramatic, cause of redistribution-induced hypokalemia is *hypokalemic periodic paralysis.* In this condition, acute and severe hypokalemia resulting from redistribution leads to flaccid paralysis or severe muscular weakness. It occurs typically during the night or the early morning, or after a carbohydrate-rich meal, and can persist for 6 to 24 hours. A genetic defect in a dihydropyridine-sensitive calcium channel has been identified in some patients, whereas other cases are associated with hyperthyroidism. Aggressive potassium administration during the attack is necessary to speed recovery and avoid the risk of respiratory failure.

Nonrenal Potassium Loss

The skin and the GI tract excrete small amounts of potassium under normal circumstances. Occasionally, excessive sweating or chronic diarrhea results in substantial potassium loss and leads to hypokalemia.[33] Vomiting or nasogastric suction may also result in loss of potassium, although gastric fluids typically contain only 5 to 8 mmol/L of potassium. The concomitant metabolic alkalosis, however, can increase urinary potassium loss and contribute to development of hypokalemia.[34]

Renal Potassium Loss

The most common cause of hypokalemia is urinary potassium loss, which typically results from drugs, endogenous hormone production, or, more rarely, intrinsic kidney defects.

Drugs. Both thiazide and loop diuretics increase urinary potassium excretion, and the incidence of diuretic-induced hypokalemia is related to both dose and treatment duration. When loop and thiazide diuretics are dosed to produce similar effects on sodium excretion, thiazide diuretics have greater effects on urinary potassium and are more likely to lead to hypokalemia. Some penicillin analogs, such as piperacillin/tazobactam, increase distal tubular delivery of a nonreabsorbable anion, which obligates the presence of a cation such as potassium and thereby increases urinary potassium excretion.[35] The antifungal agent amphotericin B directly increases collecting duct potassium secretion. Aminoglycosides may cause hypokalemia either with or without simultaneous nephrotoxicity. The mechanism is incompletely understood but may relate to magnesium depletion (see later discussion). Cisplatin is an antineoplastic agent that can induce hypokalemia from renal potassium wasting; the increased potassium excretion may persist after discontinuation of the medication. Toluene exposure, from sniffing certain glues, can cause potassium wasting, leading to hypokalemia.[36] In addition, certain herbal products, including herbal cough mixtures, licorice tea, licorice root, and gan cao, contain glycyrrhizic and glycyrrhetinic acids, which have mineralocorticoid-like effects.[37]

Endogenous hormones. Aldosterone is an important hormone regulating total body potassium homeostasis. Aldosterone predominantly causes hypokalemia by stimulating cellular potassium uptake,[2] but it can also lead to inappropriately high urinary potassium excretion during, and likely contributing to, hypokalemia.

Genetic causes. Genetic defects leading to excessive aldosterone production and resulting in hypokalemia is occasionally seen (see Chapter 49). These conditions include glucocorticoid-remediable aldosteronism, congenital adrenal hyperplasia, and apparent mineralocorticoid excess. These conditions are discussed in detail in Chapter 49.

Magnesium depletion. Magnesium deficiency can cause inappropriately high potassium excretion despite hypokalemia.[38] This appears to occur because intracellular magnesium inhibits the apical ROMK channels that are critical for distal nephron K^+ secretion. In magnesium deficiency, decreased intracellular magnesium reduces the magnesium-mediated inhibition of ROMK channels, leading to increased ROMK-mediated K^+ secretion and thereby to increased net K^+ secretion.[39] Magnesium deficiency occurs most often as a complication of prolonged diuretic use or from proton pump inhibitor (PPI) therapy. It may also result from ethanol abuse, uncontrolled diabetes mellitus, and after solid organ transplant. Magnesium deficiency should be suspected when potassium replacement does not correct hypokalemia; treatment of the underlying cause of hypomagnesemia, such as discontinuation of PPI agents, generally reverses the potassium wasting.

Primary renal defect. Intrinsic renal potassium transport defects leading to hypokalemia are rare. They include *Bartter syndrome* and *Gitelman syndrome,* which have clinical phenotypes similar to those that follow use of loop and thiazide diuretics, respectively, and *Liddle syndrome.* These are discussed in Chapter 49.

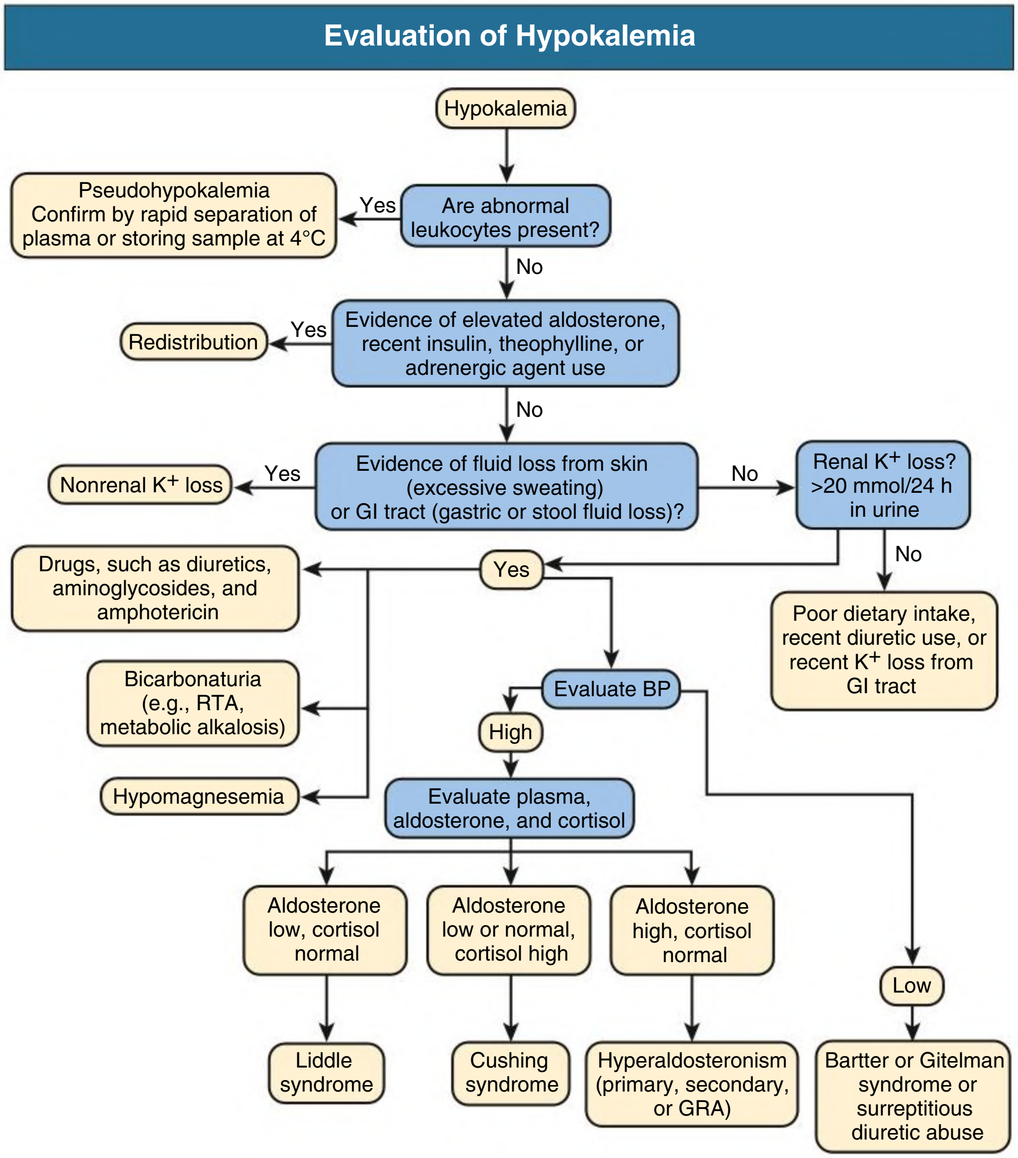

Fig. 10.4 Diagnostic evaluation of hypokalemia. *BP*, Blood pressure; *GI*, gastrointestinal; *GRA*, glucocorticoid-remediable aldosteronism; *RTA*, renal tubular acidosis.

Bicarbonaturia. Bicarbonaturia can result from metabolic alkalosis, distal renal tubular acidosis, or treatment of proximal renal tubular acidosis. In each case, the increased distal tubular bicarbonate delivery increases potassium secretion.

Diagnostic Evaluation

The evaluation of hypokalemia is summarized in Fig. 10.4. The clinician should first exclude pseudohypokalemia or potassium redistribution from the extracellular to the intracellular space. Insulin, aldosterone, fludrocortisone, and sympathomimetic agents, such as theophylline and β_2-adrenoceptor agonists, are common causes of potassium redistribution. In the hypertensive patient, hypokalemia in the absence of diuretic use suggests primary aldosteronism diuretic use suggests primary aldosteronis, for which specific testing is appropriate (see Chapter 49).

If neither pseudohypokalemia nor potassium redistribution is present, hypokalemia indicates that total body potassium depletion caused by either renal, GI, or skin losses is present. Urinary potassium loss is caused most often by diuretics. Hypomagnesemia can also cause potassium wasting and frequently results from loop or thiazide diuretic use. Less common causes of renal potassium loss include proximal and distal RTA, diabetic ketoacidosis, and ureterosigmoidostomy. Primary aldosteronism should be considered in the patient who also has hypertension. Less commonly, surreptitious diuretic use or vomiting, laxative abuse, and genetic causes such as Bartter or Gitelman syndrome should be considered when the cause of the hypokalemia is not obvious. Excessive potassium loss may also result from skin losses from excessive sweating or from the GI tract from diarrhea, vomiting, nasogastric suction, or GI fistula. Occasionally, patients are reluctant to admit to self-induced diarrhea, and the diagnosis may need to be confirmed by direct testing of the stool for cathartic agents.

Treatment

Primary short-term risks of hypokalemia are cardiovascular arrhythmias and neuromuscular weakness. Overly rapid therapy can cause acute hyperkalemia, which can cause ventricular fibrillation and sudden death. The rate of K replacement should balance these risks.

In the great majority of hypokalemic patients, emergency therapy is not necessary. The body responds to potassium losses by shifting potassium from ICF to ECF, thereby minimizing the change in extracellular

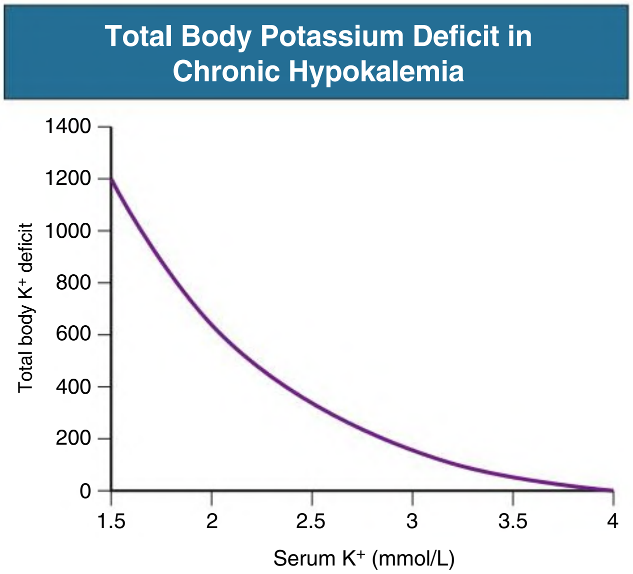

Fig. 10.5 Total Body Potassium Deficit in Hypokalemia. Because of shift of potassium from the intracellular fluid to the extracellular fluid compartment during chronic potassium depletion, the magnitude of deficiency can be masked and is generally much larger than would be calculated solely from the extracellular fluid volume volume and the change in serum potassium.

$[K^+]$. During potassium replacement, there is shift of potassium back into ICF. Consequently, the amount of potassium replacement needed is much greater than predicted by the change in extracellular $[K^+]$ and the ECF volume (Fig. 10.5).

Oral or enteral potassium administration is preferred if the patient can take oral medication and has normal GI tract function. Acute hyperkalemia is highly unusual when potassium is given orally. This reflects several factors, most prominently gut sensors that minimize changes in serum potassium levels. When potassium is given intravenously, acute hyperkalemia can occur if the administration rate is too rapid and, if it occurs, can cause sudden cardiac death. Thus, unless immediate life-threatening conditions are present, the maximum rate of IV KCl administration should not exceed 10 mmol/h. In life-threatening conditions, such as hypokalemia leading to respiratory compromise, severe hypokalemia in a patient requiring urgent surgery, and the patient with an acute myocardial infarction and life-threatening ventricular ectopy, more rapid administration may be needed. In these patients, potassium chloride (KCl) can be administered intravenously at a dose of 5 to 10 mmol over 15 to 20 minutes. Continuous ECG monitoring should be used under these circumstances. This dose can be repeated as needed.

The choice of the parenteral fluid used to administer the potassium can affect the response. In patients without diabetes mellitus, dextrose-containing fluids lead to a reflex increase in serum insulin levels, which can cause redistribution of potassium from ECF to ICF. As a result, administering KCl in dextrose-containing solutions, such as dextrose 5% in water (D5W), can stimulate cellular potassium uptake to an extent that results in paradoxical worsening of the hypokalemia.[40] Consequently, parenteral KCl should be administered in dextrose-free solutions.

The underlying condition should be treated whenever possible. If patients with diuretic-induced hypokalemia require ongoing diuretic administration, the addition of potassium-sparing diuretics may be considered. When oral replacement therapy is required, KCl is the preferred drug in most patients, except those with metabolic acidosis, in whom potassium citrate may be considered as a concomitant alkali source. Hypomagnesemia, if present, should be treated; otherwise, there may be ongoing urinary potassium losses that delay or preclude

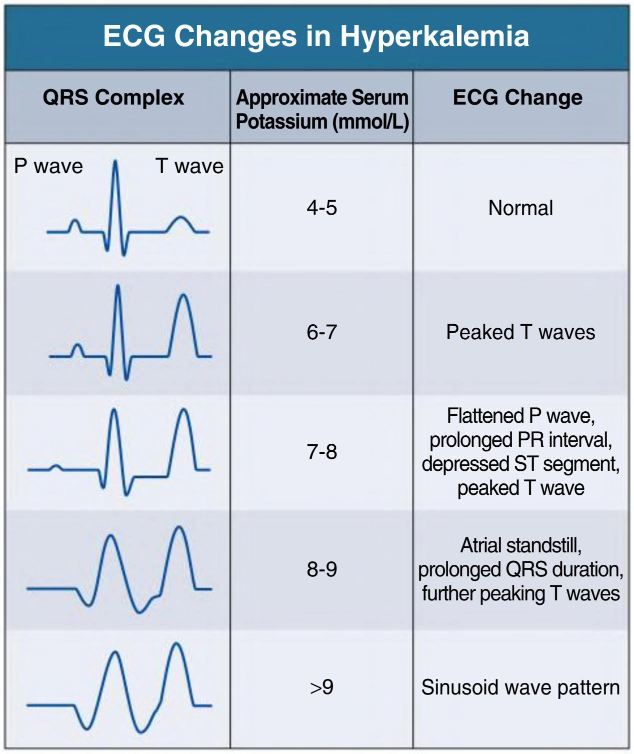

Fig. 10.6 Electrocardiographic (ECG) Changes in Hyperkalemia. Progressive hyperkalemia results in identifiable changes in the electrocardiogram. These include peaking of the T wave, flattening of the P wave, prolongation of the PR interval, depression of the ST segment, prolongation of the QRS complex, and, eventually, progression to a sine wave pattern. Ventricular fibrillation may occur at any time during this ECG progression.

correction of the hypokalemia. If clinically indicated for other reasons, the use of β-blockers, ACE inhibitors, or ARBs can assist in maintaining serum potassium levels.

HYPERKALEMIA

Epidemiology

Hyperkalemia is distinctly unusual in healthy individuals with normal kidney function not being treated with drugs that alter K^+ handling, with less than 1% of normal healthy adults developing hyperkalemia. This low frequency is a testament to potent renal mechanisms for potassium excretion. Because of this, chronic hyperkalemia, if not caused by pseudohyperkalemia or redistribution, should strongly suggest impaired renal potassium excretion, whether from decreased nephron/collecting duct number, from medications that decrease renal potassium excretion, or from adrenal insufficiency.

Clinical Manifestations

Hyperkalemia can be asymptomatic, can cause mild symptoms, or can be life threatening. Importantly, the mortality risk of hyperkalemia is independent of the patient's clinical symptoms and reflects acute effects of hyperkalemia on cardiac conduction and repolarization. This is demonstrable on the ECG (Fig. 10.6). The initial effect of hyperkalemia is a generalized increase in the height of the T waves, most evident in the precordial leads, but typically present in all leads, which is known as "tenting." More severe hyperkalemia is associated with delayed electrical conduction, resulting in an increased PR interval and a widened QRS complex. This is followed by progressive flattening and eventual absence of the P waves. Under extreme conditions, the QRS

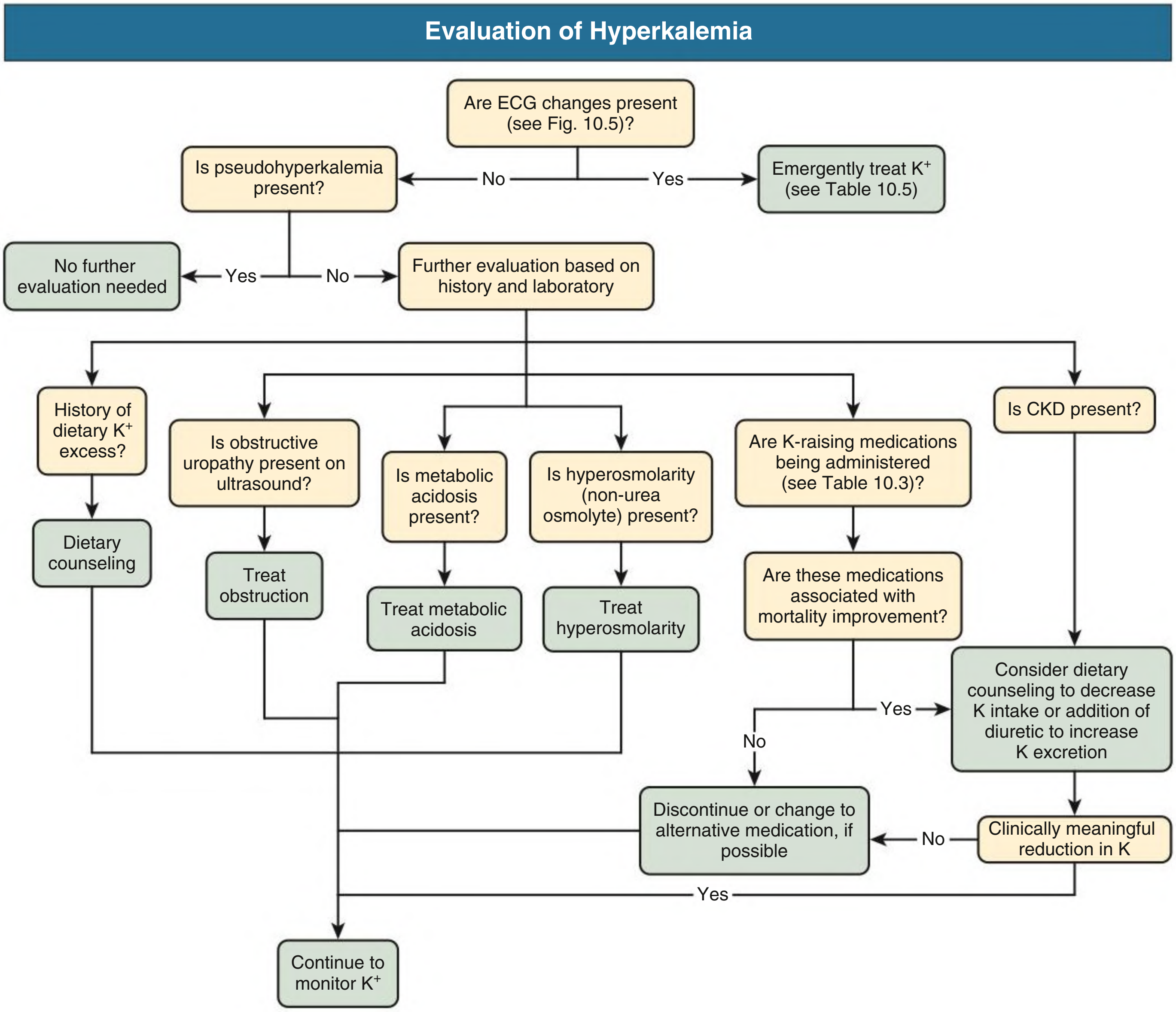

Fig. 10.7 Workup of hyperkalemia. *CKD*, Chronic kidney disease; *ECG*, electrocardiography.

complex widens sufficiently that it merges with the T wave, resulting in a junctional rhythm that degenerates to a sine wave pattern. Finally, ventricular fibrillation develops. Although the ECG findings correlate generally with the degree of hyperkalemia, the rate of progression from mild to severe cardiac effects can be unpredictable and may not correlate well with changes in the serum potassium concentration.

Hyperkalemia also has effects on noncardiac tissues. Skeletal muscle cells are particularly sensitive to hyperkalemia, causing generalized weakness. In patients with severe hyperkalemia, diaphragmatic muscle weakness may lead to respiratory failure.

Mortality

Several clinical studies have shown that chronic hyperkalemia is independently associated with increased mortality.[19–22,41–44] Whether correction of the abnormal K^+ alters the associated mortality is unknown. However, the relatively recent availability of generally well-tolerated enteric K^+-binding resins enables pharmacologic therapy of hyperkalemia without having to discontinue medications that have beneficial effects on mortality, such as ACE inhibitors/ARB, β-blockers, and MRBs, in specific clinical conditions but that can also increase potassium levels. Future studies are needed to determine whether correction of hyperkalemia with enteric K^+-binding medications will improve mortality.

Etiology

Hyperkalemia can result from pseudohyperkalemia, potassium redistribution from intracellular to extracellular space, or imbalances between potassium intake and renal potassium excretion. A diagnostic approach is shown in Fig. 10.7.

Pseudohyperkalemia

Pseudohyperkalemia refers to the condition where potassium release from blood cells occurs after the phlebotomy procedure. This most commonly results from damaged erythrocytes and is identified clinically by the presence of free hemoglobin in the plasma, reported as "hemolysis" by most clinical laboratories. If hemolysis is present, the reported serum $[K^+]$ measurement does not accurately reflect the actual serum $[K^+]$. Treatment should not be prescribed based on this value, and repeat measurement is necessary. Ischemia from prolonged tourniquet time or from exercise of the limb in the presence of a tourniquet can also lead to abnormally increased potassium values. Potassium can also be released from the other cellular elements present in blood during clotting. This can occur in patients with severe leukocytosis (>70,000/cm^3) or thrombocytosis. About one-third of patients with platelet counts of 500 to 1000 × 10^9/L exhibit pseudohyperkalemia.

Pseudohyperkalemia is diagnosed by showing that the serum [K^+] is more than 0.3 mmol/L higher than in a simultaneous plasma sample. If not caused by hemolysis, future potassium levels may need to be measured in plasma samples to allow for accurate measurement of extracellular [K^+].

Redistribution

Redistribution of potassium from ICF to ECF may occur with severe hyperglycemia (from development of hyperosmolarity), during insulin deficiency, and with administration of drugs that alter cellular K^+ uptake, such as β_2-adrenoceptor blockers, ACE inhibitors, ARB, and MR blockers. During diabetic ketoacidosis, both the hyperosmolarity that results from the hyperglycemia and the insulin deficiency contribute to transcellular K^+ shifts that are the primary cause of the often-observed hyperkalemia. During treatment with insulin, resolution both of the hyperglycemia, and its resultant hyperosmolarity, and of the insulin deficiency leads to increased cellular potassium uptake and a decrease in the serum potassium. In the patient with diabetic ketoacidosis who presents with a normal serum potassium, potassium redistribution from the insulin deficiency and hyperglycemia-induced hyperosmolarity may be masking substantial total body potassium deficiency resulting from hyperglycemia-induced polyuria. In this case, severe hypokalemia may develop during insulin treatment. Close and careful management of serum potassium may be needed. Patients who have received mannitol may also develop hyperosmolarity-induced hyperkalemia. Digoxin overdose can block cellular potassium uptake and lead to hyperkalemia that requires rapid treatment.

Excess Intake

Excessive potassium ingestion generally does not lead to chronic hyperkalemia unless other contributing factors are present. Under normal conditions, the kidney has the capacity to excrete several multiples of the mean daily potassium intake. However, if renal potassium excretion is impaired, as from drugs, acute kidney injury (AKI), or CKD, excessive potassium intake can contribute to the development of hyperkalemia.

Common sources of excess potassium intake are potassium supplements, salt substitutes, enteral nutrition products, and several common foods. As many as 4% of patients receiving potassium supplements develop hyperkalemia. "Salt substitutes" have an average of 10 to 13 mmol K/g. Many enteral nutrition products contain at least 40 mmol/L KCl; administration of 100 mL/h of such products can result in a potassium intake of about 100 mmol/day. Also, many food products are particularly high in potassium (see Table 10.1).

Impaired Renal Potassium Excretion

Chronic hyperkalemia typically involves a component of impaired renal potassium excretion. As previously discussed, renal potassium excretion is determined primarily by collecting duct potassium secretion. The number of collecting duct segments parallels GFR, and therefore impaired GFR, whether from CKD or AKI, results in impaired capacity to excrete potassium. In CKD, adaptive increases in the ability of each collecting duct segment to secrete potassium may allow renal potassium excretion to remain moderately well preserved until GFR is reduced to 10 to 20 mL/min. Multiple drugs affect renal potassium secretion (see Table 10.3); these drugs are sometimes used in combination, which further exacerbates the risk of hyperkalemia.

Obstructive uropathy leads frequently to hyperkalemia, at least in part from decreased collecting duct ENaC and Na^+,K^+-ATPase expression and activity. Hyperkalemia may persist after relief of the obstruction.[45] This impairment appears to be related to a persistent defect in collecting duct K^+ secretion and not to aldosterone deficiency.[45]

Mineralocorticoid hormones are necessary for the normal response to hyperkalemia. Lack of these hormones both causes potassium redistribution from ICF to ECF and reduces the maximal ability of the kidneys to secrete potassium. Primary adrenal insufficiency should be strongly considered in the patient with chronic hyperkalemia and other findings suggestive of adrenal insufficiency, such as spontaneous hypotension and hyponatremia.

The colon can excrete potassium, but adaptive changes in enteric potassium excretion are quantitatively small and generally are not sufficient to maintain normal potassium homeostasis.

A rare genetic disorder, pseudohypoaldosteronism type 2 (PHA2; also known as Gordon syndrome) is characterized by hypertension, hyperkalemia, non–anion gap metabolic acidosis, and normal GFR[46] and is discussed in Chapter 49.

Determining the Role of Excessive Potassium Intake in Chronic Hyperkalemia

In most patients, a careful history and measurement of K^+ in a 24-hour urine collection will identify the role of excessive dietary potassium intake in the development of chronic hyperkalemia. In selected patients, specifically those not using diuretics and either unable to be compliant with or preferring not to perform a 24-hour urine collection, assessment of the urine potassium/creatinine ratio in a random urine specimen may be used, with urinary potassium greater than 60 mmol K^+/g creatinine suggesting excessive dietary K^+ intake is present. If there is a clinical concern about possible hypoaldosteronism as a primary cause of the hyperkalemia, such as in a patient with concomitant normal or low blood pressure and hyponatremia, the transtubular K^+ gradient (TTKG), when considered in context with assessment of K^+ intake, can be helpful (Table 10.4).

Treatment

Acute Therapy

Acute therapies for hyperkalemia are divided into those that minimize the cardiac effects of hyperkalemia, those that induce potassium uptake by cells resulting in a decrease in serum potassium, and those that remove potassium from the body (Table 10.5). Sodium bicarbonate ($NaHCO_3$) therapy should not be used unless the patient is frankly acidotic (pH < 7.2) or there is substantial endogenous renal function. Hypertonic $NaHCO_3$ therapy (e.g., 50 mmol in 50 mL of sterile water) can worsen intravascular volume overload, as frequently seen in the patient with oliguric AKI; can cause acute hypernatremia; and can acutely *increase* serum potassium in the anephric patient, likely as a result of the acute increase in plasma osmolality.[47]

Blocking cardiac effects. IV calcium administration rapidly antagonizes the effects of hyperkalemia on the myocardial conduction system and on myocardial repolarization and does so without changing plasma K^+. Calcium should be given intravenously as the initial therapy if unambiguous ECG changes of hyperkalemia are present. If the ECG is ambiguous, comparison with a previous ECG may be helpful. Patients with a prolonged PR interval, a widened QRS complex, or the absence of P waves should receive IV calcium in the form of either calcium chloride or calcium gluconate without delay. Responses typically occur within 1 to 3 minutes but generally last for only 20 to 60 minutes. Doses may be repeated as needed if ECG changes persist or they recur. If a delay in more definitive therapy, such as institution of dialysis, is anticipated, a continuous calcium infusion can be used.

IV calcium is relatively safe if certain precautions are taken. IV calcium should not be administered in $NaHCO_3$-containing solutions because calcium carbonate ($CaCO_3$) precipitation can occur. Hypercalcemia, which occurs during rapid calcium infusion, can potentiate the myocardial toxicity of digoxin. Thus, patients taking digoxin, particularly if they have evidence of digoxin toxicity as a contributing cause of hyperkalemia, should be given calcium as a slow infusion over 20 to 30 minutes.

Cellular potassium uptake. The second most rapid way to treat hyperkalemia is to stimulate cellular potassium uptake using either insulin or β_2-adrenergic agonist administration. Insulin rapidly stimulates cellular potassium uptake and should be administered intravenously to ensure rapid and predictable bioavailability. The effect of insulin on serum [K^+] is seen generally within 10 to 20 minutes and can last for 4 to 6 hours. Glucose is generally coadministered to avoid hypoglycemia but may not be needed if hyperglycemia coexists. This is particularly important because extracellular glucose in patients with diabetes mellitus can function as an "ineffective osmole" and can increase the serum potassium concentration. It is important to recognize that decreased kidney function can lead to delayed insulin clearance, and hypoglycemia can result from IV insulin administration, even if glucose is coadministered, because glucose uptake may occur more rapidly than insulin clearance. Accordingly, patients given IV insulin for treatment of hyperkalemia should be closely monitored for the development of hypoglycemia. If dialysis is indicated and a delay in its initiation is anticipated, administering a continuous infusion of insulin, 4 to 10 U/h (with 10% dextrose in water, $D_{10}W$), may be beneficial; periodic monitoring of serum glucose and potassium is required.

β_2-Adrenoceptor agonists directly stimulate cellular potassium uptake and can be administered via IV, subcutaneously (SC), or by inhalation. However, β_2-agonist therapy frequently causes tachycardia, and as many as 25% of patients do not respond to β_2-agonist therapy given by nebulizer.[48] A common mistake when administering nebulized β_2-adrenoceptor agonists is underdosage; the dose required is two to eight times that usually given for bronchodilation and is 50 to 100 times greater than the dose administered by metered dose inhalers.

Potassium removal. Most patients with persistent, severe hyperkalemia benefit from K^+ removal from the ECF. Definitive treatment of these patients requires potassium elimination through the kidneys, GI tract, or dialysis. Patients with chronic hyperkalemia may also benefit from drugs that stimulate renal or stool K^+ excretion. With chronic or mild hyperkalemia, loop or thiazide diuretics increase renal potassium excretion; this is particularly important for patients with hyperkalemic renal tubular acidosis (Type 4 RTA), in whom the hyperkalemia is an important causative factor in the development of the metabolic acidosis.[49,50] With life-threatening hyperkalemia, diuretics should generally not be used because the rate of renal potassium excretion is not adequate. If a rapidly reversible cause of kidney failure is identified (e.g., obstructive uropathy or prerenal azotemia and kidney failure from volume depletion), treating the underlying condition may be sufficient, as long as there is close observation of serum potassium and continuous ECG monitoring.

A second mode of potassium elimination is with enteric K^+-binding medications. Cation exchange resins, such as sodium polystyrene sulfonate or calcium polystyrene sulfonate (calcium resonium), have been used for decades. These resins exchange sodium or calcium, respectively, for potassium in the GI tract, enabling potassium elimination. They can be administered orally or as a retention enema. The rate of potassium removal is relatively slow, requiring about 4 hours for onset, although

TABLE 10.4 Transtubular Potassium Gradient

The transtubular potassium gradient (TTKG) is a measurement of net K^+ secretion by the collecting duct after correcting for changes in urinary osmolality and is often used to determine whether hyperkalemia is caused by aldosterone deficiency/resistance or whether the hyperkalemia is secondary to nonrenal causes. As with all diagnostic aids, clinical correlation is indicated, and potassium intake should be assessed.

$$TTKG = \frac{[K^+]_U / [K^+]_S}{Osmo_U / Osmo_S}$$

where $[K^+]_U$ and $[K^+]_S$ are the concentration of K^+ in urine and serum, respectively, and $Osmo_U$ and $Osmo_S$ are the osmolality of urine and serum, respectively.

TTKG Value	Indication
6–12	Normal
>10	Suggests normal aldosterone action and extrarenal cause of hyperkalemia
<5–7	Suggests aldosterone deficiency or resistance
After 0.05 mg 9α-Fludrocortisone:	
>10	Hypoaldosteronism is likely.
No change	Suggests a renal tubule defect from either K^+-sparing diuretics (amiloride, triamterene, spironolactone), aldosterone resistance (interstitial renal disease, sickle cell disease, urinary tract obstruction, PHA1), or increased distal K^+ reabsorption (PHA2, urinary tract obstruction)

TABLE 10.5 Acute Treatment of Hyperkalemia

Mechanism	Therapy	Dose	Onset	Duration
Antagonize membrane effects	Calcium	Calcium gluconate, 10% solution, 10 mL IV over 10 min	1–3 min	30–60 min
Cellular potassium uptake	Insulin	Regular insulin, 10 U IV, with dextrose 50%, 50 mL, if plasma glucose <250 mg/dL	30 min	4–h
	β_2-Adrenergic agonist	Nebulized albuterol, 10 mg	30 min	2–h
Potassium removal	Sodium polystyrene sulfonate or calcium polystyrene sulfonate (calcium resonium)	30–60 g PO in 20% sorbitol *or* 30–60 g in water, per retention enema	4–6 h	6–12 h
	Patiromer	16.8–25.2 g	4–6 h	6–12 h
	Sodium zirconium cyclosilicate	10 g	4–6 h	6–12 h
	Hemodialysis	—	Immediate	Until dialysis completed

administering the resin as a retention enema results in more rapid onset of action. When given orally, cation exchange resins are generally administered with 20% sorbitol to avoid constipation. If given as an enema, sorbitol should be avoided because rectal administration of cation exchange resins with sorbitol may increase the risk of colonic perforation.[51] Questions have been raised recently as to the efficacy of these compounds and whether the risk of colonic perforation exceeds their benefits.[52] Two new enteric potassium-binding medications, patiromer and sodium zirconium cyclosilicate,[53,54] have been developed. They have a better subjective side effect profile than the K^+-binding resins, which leads to better long-term tolerability. Although all enteric K^+-binding medications appear to lower K^+ levels within 6 to 24 hours, none should be relied on for the treatment of life-threatening hyperkalemia.

Acute hemodialysis is the primary method of potassium removal in AKI or advanced CKD when hyperkalemia is life-threatening. Serum potassium can decrease as much as 1.2 to 1.5 mmol/h with a low potassium (2 mmol/L) dialysate. In general, the more severe the hyperkalemia, the more rapid should be the reduction in serum potassium until K^+ is less than 6.0 mEq/L. However, care should be taken to avoid reducing the serum potassium too rapidly in patients with coronary heart disease or severe cardiac arrhythmias. In these patients, longer dialysis with dialysate potassium of 3 mmol/L allows serum potassium to equilibrate to that level. However, rigorous data supporting which patients should be treated with higher K^+ dialysate are not available. Our policy is to use 3 mmol/L K^+ dialysate in patients with a history of sudden cardiac death. We also recommend rechecking serum potassium after completion of the dialysis procedure to verify effective correction of the hyperkalemia, waiting 1 to 2 hours to allow transcellular reequilibration. Continuous dialysis modalities, such as peritoneal dialysis and continuous venovenous hemodialysis, generally do not remove potassium sufficiently quickly for use in patients with life-threatening hyperkalemia but may be justified under unusual circumstances where hemodialysis is not available.

If dialysis is delayed, as when access to equipment or nursing support is not immediate, or while vascular access is being established, other therapies should be instituted and continued until hemodialysis is begun.

Specific therapies are available for certain causes of hyperkalemia. For example, digoxin-specific Fab fragments are beneficial in patients with severe digitalis glycoside toxicity.[55] Hyperkalemia in patients with acute urinary tract obstruction may be treated by relieving the obstruction, but the rate of potassium excretion afterward is variable, and frequent measurement of serum potassium is necessary.

Chronic Treatment

Management of chronic hyperkalemia is a common and often challenging problem, particularly in the CKD patient. Patients with CKD have impaired capacity to excrete a potassium load rapidly and are often treated with medications, such as ACE inhibitors, ARB, β-blockers, and MRBs, that can cause hyperkalemia.

Although most patients with chronic hyperkalemia are receiving medicines associated with the development of hyperkalemia, discontinuing all drugs that can cause hyperkalemia may not be the correct approach. Many medicines, such as ACE inhibitors, ARBs, β-blockers, and MRBs, have significant cardiovascular, renoprotective, and mortality benefits. Discontinuing these drugs simply because of mild hyperkalemia is therefore not recommended in the first-line management. However, if the patient is receiving combined therapy with ACE inhibitor and ARB, discontinuing one of these drugs may decrease the risk of further hyperkalemia, and this does not appear to be associated with adverse renal effects. Similarly, if a patient is receiving combination therapy of either an ACE inhibitor or an ARB with a direct renin inhibitor, such as aliskiren, the direct renin inhibitor may be discontinued.

Instead of immediately discontinuing such beneficial drugs without review, a careful assessment should be made for medicines that cause hyperkalemia but are not necessary for renoprotective or cardioprotective benefits and therefore can be discontinued, such as NSAIDs, cyclooxygenase (COX)-2 inhibitors, potassium-sparing diuretics (e.g., amiloride, triamterene), and oral KCl or potassium citrate supplementation. If hyperkalemia persists, addition of diuretics to increase potassium excretion can be considered. Thiazide diuretics may be preferred because, when adjusted for their effect on sodium excretion, thiazide diuretics are associated with a greater increase in renal potassium excretion than loop diuretics. In the patient with CKD, the thiazide diuretic metolazone may be effective.[56] Combination of thiazide and loop diuretics is more effective than either alone. Finally, screening for and treating metabolic acidosis may facilitate correction of the hyperkalemia. Alkali therapy may also increase K^+ clearance.

A dietary history should be obtained, and if the patient is ingesting a diet with potassium-rich foods, instruction on avoidance of these foods should be provided. However, routine instruction in a low-potassium diet for CKD patients is not recommended. Recent studies have suggested that a diet higher in potassium-rich foods may be associated with slower progression of CKD.[57]

Finally, chronic therapy with drugs that increase enteric K^+ excretion can be considered, such as patiromer or sodium zirconium cyclosilicate.[53,54] These drugs may allow for greater use of drugs that improve mortality in selected patient populations but whose use is otherwise limited because of development of hyperkalemia (e.g., ACE inhibitors and ARBs).

Synthetic mineralocorticoid therapy, such as fludrocortisone, has also been used for the treatment of chronic hyperkalemia. This approach may be helpful in patients with chronic hypotension caused by adrenal insufficiency who have normal kidney function. In CKD patients, the accompanying renal sodium retention, intravascular volume expansion, and increased blood pressure are relative contraindications to synthetic mineralocorticoids. Furthermore, selective blockade of the mineralocorticoid receptor appears to decrease kidney injury in several experimental models of renal injury, suggesting that administration of synthetic mineralocorticoids may be injurious. As a result, their adverse effects may exceed their benefits in patients with CKD.

SELF-ASSESSMENT QUESTIONS

1. Deficiency of which of the following ions can result in renal potassium wasting?
   A. Phosphate
   B. Calcium
   C. Sulfate
   D. Magnesium
2. Which of the following drugs is not associated with development of hyperkalemia?
   A. Terazosin
   B. Propranolol
   C. Spironolactone
   D. Cyclosporine
3. Hypokalemia is associated with which of the following conditions?
   A. Decreased insulin sensitivity
   B. Increased renal interstitial fibrosis
   C. Increased risk of hepatic encephalopathy
   D. Polyuria
   E. All of the above

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